Darshan Daga

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About me

Generative AI and Data Science Specialist with 8+ years of experience delivering end-to-end AI solutions, specializing in LLMs, prompt engineering, and Retrieval-Augmented Generation (RAG). Proven expertise in deploying NLP applications, integrating open-source models, and building scalable RESTful APIs using FastAPI. Having experience translating complex business problems into actionable insights using Python, SQL, PySpark, and PowerBI. Strong background in healthcare, manufacturing, telecom and tech domains with a track record of delivering high-impact data products and driving revenue growth.

Career objective

To be part of a challenging team which strives for the better growth of organization and which explores my potential and provides me with the opportunity to enhance my talent with an intention to be an asset to the company.

Education

B.E., Ramdeobaba college of engineering and management, Computer Science and Engineering — May 2012 to June 2016

PGP, Great Lakes, Machine learning — Jan 2019 to Sept 2019

Experience

Publicis Sapient, Senior Associate L2 - Data science — Pune, India

May 2024 to Present
1 year

- Engineered an NL-to-SQL conversion system enabling non-technical users to query databases using natural language, eliminating SQL knowledge barriers while maintaining data governance.
- Implemented Retrieval-Augmented Generation (RAG) to enhance response accuracy and contextual relevance
- Integrated knowledge graphs to establish semantic relationships between data entities, improving query understanding and result relevance.
- Leveraged agentic ai frameworks including LangGraph agents and AutoGen to create a robust, multi-step reasoning pipeline for complex query interpretation.
- Built high-performance REST APIs using FastAPI to handle query processing and data retrieval with efficient validation and documentation

MathCo, Senior Associate - Data science — Pune, India

Oct 2021 to May 2024
2 years 7 months

- Developed and implemented advanced methods and Generative AI algorithms using Python, leveraging open source LLMs, to analyze data coming from the website of a well known manufacturing company, achieving actionable insights and predictive capabilities, helping user navigate better in Their buying journey and FAQs.
- Spearheaded the integration of Generative AI techniques, including Retriever Augmented Generation and Vector Search, within the AWS platform, leveraging its AWS bedrock service, to optimize information retrieval and enhance data-driven decision-making processes.
- Translated 120 user reports into a consolidated PowerBI dashboard, improving reporting efficiency by 40 percent.
- Identified pain points, resulting in tailored solutions that reduced reporting time by 50 percent.
- Collaborated with the Associate Director of US oncology portfolios to develop ad hoc reports tracking critical KPIs such as market penetration, enrollment into different services, access and reimbursement, Length of therapy, persistence, adherence, etc.
- Worked on designing *test and learn campaign* that primarily worked on the measuring the strategy change.
- Helped the team build a *stratification model* that segregated the group in test split and the learn split.
- *Derived metrics* that measured the call pickup rates of the campaign and found out if the lift in the campaign is significant or not.
- Implemented *propensity score matching algorithm* to figured out control split.
- Lead a team to *develop a EDA capability* specific to the campaign data. The results of which were presented on PowerBI.
- Worked on *launch analytics of a drug* to understand the market's response by strategising and tracking KPI's.
- Helped the team build a *Power BI dashboard* that tracks and analyses the performance of the Patient Support program to help client make necessary changes by identifying the weak points of the program.
- The result of this was an increased revenue of *500,000 dollars*.
- Worked on time series *forecasting models like ARIMA and SARIMA* that predict the potential enrollments in the upcoming 180 days in the Patient Support Program.
- *Lead the team of 5 analysts* and helped them convert funnel queries to *pySpark* code.

Kepler CP - Tech Mahindra, Data scientist

Pune, India
Aug 2016 to Oct 2021
5 years 2 months

- Worked on *prediction and analytics* of demand and supply of technicians using *recommendation engine.*
- Responsible for developing models that manages technicians optimally and suggests the manager how and which area to use them in.
- Wrote *wrapper classes* in python to pre-process data that needs to be fed in the models.
- Implemented *greedy searches and Predictive models* as a part of the project.
- Was a part of End to end *CI/CD deployment pipeline using Jenkins, Docker, Kubernetes* and making a *flask API* for our application.
- Responsible for *developing 6-week average script* that takes historical 6 week data and predicts the customer churn for the next 90 days.
- Responsible for *dockerizing* the entire application and *deploying it on Azure*
- Worked on a *Natural language processing(NLP)* project which included reading an excel that had fields like acronym and its description.
- Using NLP algorithms we had to identify which bucket the application acronym would fit in and which category does the acronym belong to.
- Responsible for *importing data, preparing stop words, removing newline characters, tokenizing keywords and cleaning the text.*
- Creating *bigram and trigram models, bigrams and lemmatize.*
- Creating *dictionary and corpus* needed for *topic modeling*.
- Building topic models and view topics in *LDA models* and topics keywords.

Projects

Projects done as a part of post-graduate program in machine learning - Great Lakes

Jan. 2019 - Sept. 2019

- Performed EDA on pima-Indian-diabetes dataset using Python. Functions of Pandas,Numpy , Seaborn and Matplotlib libraries to plot graphs to get insights about the dataset.
- Analyzed health insurance for insurance business. Used hypothesis testing and Visualizations to leverage customer's health information like smoking habits,BMI ,etc for checking statistical evidence and making decisions for insurance business.
- Identified potential loan customers for Thera bank using Classification techniques.Compared models built with Logistic Regression and KNN algorithm in order to select best performing one
- Diagnosing Parkinson Disease which involved using classification Algorithms and Ensemble techniques to diagnose it using the patients voice data. Used models such as Naive Bayes, Logistic Regression, SVM, Decision Tree, Random Forest and compared the accuracy.
- Classified vehicles into different types based on silhouettes which may be viewed from any angles. Used PCA in order to reduce dimensionality and SVM for classification.
- Predicting the strength of high performance concrete which involved feature exploration and selection to predict the strength. Used Regression models like Decision tree regressor to find out the most important features. Cross validation techniques and grid search were used to tune the parameters for best model performance.
- Capstone Project this project involved using features variables available in Innercity house price dataset. Different ML models like Regression, Ensemble Techniques were used to analyze and predict the house price. Also, grid search were used to tune different parameters associated with models. This application was deployed to AWS using elastic beanstalk.

Additional Experience And Awards

- Won 4 client appreciation awards in an year in Mathco.
- Won the best team award in 2023.
- Won the Pat on the back award in 2017,2019,2020 in tech mahindra.

Technologies

Tech Stack: Python, SQL, MongoDB, Pandas, matplotLib, scipy, PowerBI, genAI

Libraries: Pandas, numpy, matplotlib, scipy, NLTK, spacy, Langchain, Langgraph, huggingFace, autogen